

CHAPTER 30

OPERATIONS RESEARCH TECHNIQUES FOR ROBOTICS SYSTEMS

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1 INTRODUCTION

1.1 Scope of the Chapter

The scientific field of *operations research*, also known as *management science*, has much to offer towards the solution of complex planning problems such as those that arise in robotic systems. This chapter discusses a variety of operations research techniques, with the aim of enabling both industrial and academic readers to establish connections between those techniques and robotic systems applications. An overview of the most important techniques is provided, along with enough details to enable the reader to assess the potential relevance and applicability of the technique in each case. Many related references that give further details are provided.

The chapter is written with the expectation that the reader is either a system manager or controller involved in implementing or operating a robotic system, or a researcher familiar with operations research techniques and hoping to apply them to a robotic system in a way that would be potentially valuable to the manager or controller.

The use of operations research techniques for the planning, design, and operation of robotic systems is valuable because a robotic facility is a complex system. Even a small facility typically consists of many interconnected hardware and software components and involves the use of various limited resources such as buffer space, part feeders or orienters, and material-handling equipment. The design and operation of such a system are difficult tasks because of the many available choices and also because of the interactions between the different components. Complex choices of this type can often be evaluated and resolved more effectively by using the powerful techniques of operations research than by any other strategy. This suggests the importance of operations research as a topic in a study of industrial robotics.

Space considerations dictate the extent to which each technique can be discussed. In general, widely used operations research techniques are discussed briefly, and their use for robotic systems is illustrated with supporting references. However, where the study of robotic systems has generated important new problems or techniques, a more detailed discussion is provided.

1.2 Operations Research

Examples of the use of a scientific approach to the management of organizations can be found as far back in time as the late nineteenth century. However, it is generally agreed that the origins of operations research as a coherent body of knowledge can be traced to military support activities during World War II. A lack of critical resources during war-time, and the urgency of the war effort, motivated both U.K. and U.S. military leaders to encourage the development and use of scientific management techniques for solving complex logistical problems. To this end, teams of scientists in both countries were formed

to study strategic and tactical operations and to improve their efficiency. Operations research has been credited with making major contributions to the war effort.

After the war ended, operations research spread quickly into the private sector, where it was applied with considerable success to complex planning problems. The usefulness of operations research was greatly enhanced by significant scientific advances (for example, the development of the simplex method for linear programming by G. Dantzig in 1947). This usefulness was further enhanced by the development of high-speed digital computing, which made possible the implementation of operations research techniques for much larger and more complex planning problems in industry. By the early 1950s the use of operations research by technical specialists could be frequently observed in manufacturing and service industries as well as in government planning problems. Within universities both business and engineering schools developed programs in operations research to support the growing interest in industry. Related programs also developed in mathematics and computer science departments. Finally, the growth of personal computing and the widespread development of affordable and easy-to-use software that occurred in the 1980s and 1990s have given access to operations research techniques, not just to technical specialists but to the entire industrial and commercial communities.

A major source of information about operations research is the continuing series *Handbooks in Operations Research* (1989–). A comprehensive introductory text on operations research is provided by Hillier and Lieberman (1995). This book also provides references to classic works in each general topic area within operations research. An interesting history of the important topic area of *mathematical programming* is provided by Lenstra, Rinnooy Kan, and Schrijver (1991). For a comprehensive treatment of the operations research techniques underlying *production and manufacturing* decisions, the text by Nahmias (1993) is highly recommended. The professional society that promotes operations research in the United States is INFORMS, which has over 11,000 members and a useful homepage at “www.informs.org”. A valuable compendium of related information appears in the 1997 OR/MS Resource Directory. Finally, the definitive on-line source for operations research information, including people, educational programs, research groups, companies, and many electronic links, is the homepage of M. Trick at “<http://mat.gsia.cmu.edu>”.

1.3 Phases of the Robotic System Design Process

In the planning, implementation, and operation of a robotic system, an organization either explicitly or implicitly goes through all the following distinct phases of activity:

1. *System planning.* This involves a feasibility study and preliminary system design, usually with the consideration of several alternatives, as well as preliminary economic evaluations.
2. *System design.* This involves deciding which operations will be performed by the system, selecting equipment, and deciding how it is to be configured. This stage is critical to the eventual performance of the system, since the choices made here limit the options that are available under the later system operation phase.
3. *System operation.* The system should be operated in an efficient way that satisfies production targets and other management criteria (for example, reduction of work-in-process). It should be maintained with minimum disruption to ongoing processes. Also, the performance of the system should be regularly evaluated, and potential modifications, upgrades, and expansion should be considered.

In each of these activity phases appropriately used operations research techniques can greatly assist the decision-maker. Details of the system planning, design, and operation issues, as well as suitable operations research techniques for evaluating them, are discussed in Sections 2 through 4, respectively, of this chapter.

1.4 Choosing an Operations Research Technique

Operations research has developed and now offers a wide variety of techniques. These techniques differ considerably in their attributes and in the type of evaluative and prescriptive information they provide. Most operations research techniques construct an abstract representation, or *model*, of a decision problem, which is then solved using

computer software. There is an important general tradeoff, discussed later in this subsection, with respect to the solvability and realism of a model.

To give the reader an overview of the various available operations research techniques, we make use of a simple classification scheme based on two attributes of those techniques. The first attribute describes whether the technique is based on a *deterministic* or *probabilistic* model. In deterministic models the assumption is made that all relevant data are known with certainty. However, deterministic models can be used to develop good approximations for problems in which uncertainty is present in a limited way. For example, some redundancy or estimated “safety factor” can be built into the model to allow for uncertainty. By contrast, in probabilistic models uncertainties that arise in a problem are explicitly incorporated into a mathematical model that represents that problem. The second attribute describes whether the technique uses a *static* or *dynamic* model. Static models look at a snapshot of a system’s performance or at an aggregate measure that summarizes performance over a period of time. By contrast, dynamic techniques explicitly study the behavior of the system at various points in time. In the case of probabilistic models that use an aggregate measure of long-run system performance, the term *static* is replaced by *steady-state*.

Given these attributes, we can classify operations research techniques as shown in Table 1. In Table 1 the reader can find any technique discussed in the remainder of this chapter as well as a reference to the section where it is discussed. It is thus easy to understand how the important issues of uncertainty and time horizon can be modeled by that technique. Some techniques may be used with various attributes, in which case they are located in Table 1 according to their most typical use. However, in the case of two techniques a * denotes frequent use under two different sets of attributes.

In choosing an operations research technique an important criterion is finding the right level of complexity. More complex models have a greater ability to capture the full details of a decision problem, but they tend to be harder to solve than simpler models. The choice of an appropriate level of complexity may also depend upon what phase of the design process is being considered. For example, early stages of the design process may require only approximate comparisons of the performance of different designs. However, in planning detailed operations it may be useful to *optimize* the system, which requires a model that captures all or almost all of the complexities of the real-world problem. Such a model may have many variables and constraints, and finding optimal solutions to the model may present a difficult computational challenge that requires expensive hardware and software. Moreover, gathering data and formatting the input information may be time-consuming and costly. Therefore, the costs involved in the optimization effort should be

Table 1 Classification of Operations Research Techniques

Attribute 1	Attribute 2	
	Static/Steady-State	Dynamic
Deterministic	integer and mixed integer programming (2.3.1)	location and layout* (2.3.5)
	linear programming (2.3.2)	scheduling* (4.3.2)
	analytic hierarchy process (2.3.3)	
	heuristics (2.3.4)	
	location and layout* (2.3.5)	
	group technology and cell formation (3.3.1)	
	batching (4.3.1)	
	scheduling* (4.3.2)	
	decision analysis (2.3)	strategic planning (2.3)
	queueing models (3.3.2)	forecasting (2.3)
Probabilistic	queueing networks (3.3.3)	simulation (3.3.4)
		reliability and maintenance (3.3.5)
		hierarchical production planning (4.3.3)

weighed against the potential benefits to be gained from the use of improved operating procedures that may be identified by the model.

1.5 Structure of the Chapter

The chapter is organized according to the phases of system planning, design, and operation identified in Section 1.3. Each of the next three Sections (2 through 4) considers one of those phases. For each phase we begin with a discussion of the most important issues and requirements that need to be considered. This discussion is followed by a summary of several performance measures that might be appropriate for that phase of the overall process. Then an overview is presented of the operations research techniques that are potentially useful for problems arising in that phase, along with many references giving examples and additional information. Finally, Section 5 places the various uses of operations research techniques for robotic systems planning, design, and operation within an overall model of organizational decision-making and discusses the importance of decision support systems.

2 SYSTEM PLANNING

This section considers the preliminary system planning phase. Here the concern is to evaluate the feasibility of a robotic system by developing initial estimates of cost, staffing and training requirements, floor space requirement, return on investment, and other measures of importance to management.

2.1 Issues and Requirements

In order to establish guidelines for the scope of the planning study, the issues and requirements that are most important to management should be identified before the study is initiated. Typically those issues and requirements include the following:

1. *Management objectives.* The purposes of considering the adoption of a robotic system need to be identified. The expected benefits should be listed. It is important to estimate the costs of disruption to existing processes resulting from the new system. All parties likely to be affected by the new system must be consulted. The expected benefits of a robotic system typically include improved productivity, reduced labor costs, improved worker safety, improved product quality, and increased product flexibility. However, in any given application there may be some benefits that are predictable in advance and others that can only be evaluated once the system is fully operational. Estimates for the latter should be included in preliminary evaluation models.
2. *Location and layout.* The distinction between location and layout issues is that location relates to the placement of the plant itself, whereas layout relates to the positioning of a robotic system within an existing plant. In location problems proximity to sources and to markets is important, as are cost factors. The layout decision attempts to make material flow as efficient as possible. If a high priority placement is given to the robotic system, it is important to ensure that the flow of material to other systems within the plant is not significantly disrupted.
3. *Operations and equipment to be automated.* The operations that will be considered for automation should be identified. The use of group technology and cellular manufacturing principles should be evaluated. Robot-served cells can be organized by product (i.e., all machines that are used to produce a given product or family of products) or by process (i.e., all products processed on a given machine). While it is not necessary at the planning stage to finalize these decisions in detail, group technology principles (see Section 3.3.1) are available to develop approximate scenarios.
4. *Resource requirements.* The major resource requirements to be considered are floor space, material flow capacity, capital cost, and adequately trained personnel. In order to assess the availability of these resources, estimates of the required resources should be prepared.

The purpose of identifying these issues and requirements is to focus the preliminary design study on the decisions and options that are most important. If this is done, the study becomes much simpler and consequently the variety of operations research techniques that can be applied increases. Since different techniques provide substantially dif-

ferent evaluative and prescriptive information, it is useful to have a range of these techniques available.

Given these general guidelines, specific requirements of this preliminary phase are typically to identify the following:

1. One or more alternative systems that can meet management objectives
2. For each alternative, estimates of performance indicators
3. For each alternative, the equipment and the operations to be automated

2.2 Performance Measures

In order to make relevant comparisons between current technology and available alternatives for robotic systems, several performance measures are needed. The following are typically useful:

1. *Amount of investment.* The total capital cost and annual incremental operating cost of the new system must be estimated.
2. *Return on investment.* The return on investment, the net present value, and the payback period must be estimated. These measures and a variety of others are discussed by Canada and Sullivan (1989).
3. *Flexibility.* It is important to assess how easily short-term changes in production requirements can be handled by the new alternatives. The potential for long-run capacity expansion and for adapting to fundamental shifts in product design are relevant. Also, the impact of significant reductions in product life cycles on the system is useful to know.

It should be emphasized that these questions are relative and not absolute. That is, they should be used only in the context of a comparison between the current and future alternatives. Useful references in the area of capacity expansion include the books by Manne (1967) and Freidenfelds (1981). Models for technology acquisition are described by McCardle (1985), Monahan and Smunt (1989), and Fine and Freund (1990). A sophisticated multiple objective decision model for evaluating advanced manufacturing systems is provided by Demmel and Askin (1992). An extensive review of the literature related to the selection of industrial robots, and an overview of various modeling approaches to the selection problem, are provided by Khouja and Offodile (1994). A decision support system that combines optimization and simulation tools to evaluate a computer-integrated manufacturing system is described by Kassiech, Ravinder, and Yourstone (1997).

2.3 Operations Research Techniques for System Planning

We begin with three examples of general planning techniques that can be applied to robotic systems. Since applications of these techniques to robotic systems are similar to well-documented applications elsewhere, we provide only brief comments and several references containing additional information. We then discuss in greater detail several techniques that can have a major impact on the planning of a robotic system.

1. *Strategic Planning.* Strategic planning is usually associated with long-term decisions. The purchase and implementation of an expensive new robotic system is a long-term commitment. Valuable references on the topic of strategic planning generally include Porter (1980), Hayes and Wheelwright (1984), and Cohen and Zysman (1987). As noted by Skinner (1974), changes in the manufacturing task can be a major source of inconsistency between company policies in different functional areas. Goldhar and Jelinek (1983) discuss arguments in favor of new manufacturing technology.
2. *Forecasting.* A natural use of forecasting to aid in planning a robotic system is to predict future demand for the products being considered for automated production. This is important because the benefits of a robotic system will depend upon the production requirements for its products and because of the considerable cost of such a system. A variety of time series and other forecasting techniques are available. Useful general references for forecasting include Box and Jenkins (1970), Montgomery and Johnson (1976), and Gross and Peterson (1983). An interesting review of forecasting practices in industry is provided by Sanders and Manrodt

(1994). Yurkiewicz (1996) presents a thorough survey of commercially available forecasting software.

3. **Decision Analysis.** Decision analysis (which includes decision trees) is a technique that has direct application to robotic systems. Uncertainties involving, for example, market strength or competitors' reactions can be included in a model to evaluate which of several robotic systems to build. Other relevant uses of decision analysis involve planning where to locate a new robotic facility when the location of future demands is unknown. Good general references for decision analysis include Keeney and Raiffa (1976), Bunn (1984), and Goodwin and Wright (1991). Fishburn (1989) provides an interesting perspective on the historical development of decision theory and related topics. Mehrez, Offodile, and Ahn (1995) use decision analysis to study the impact of a robot's *repeatability*, i.e., physical precision over time, on output rate and quality. Buede (1996) presents a thorough survey of commercially available decision analysis software.

2.3.1 Integer and Mixed Integer Programming

Integer programming (Nemhauser and Wolsey, 1988; Salkin and Mathur, 1989) is an optimization technique that aids in complex decision-making. The problem-solving power of this approach is demonstrated by Johnson, Kostreva, and Suhl (1985). We illustrate the use of this technique by developing a model that will maximize the value of the output of a robotic system over a given time horizon, T . The decisions to be made include which operations to include in the robotic system and which robot-served workstations to purchase, subject to a capital investment budget. We use the following notation for input data.

- $O_1, \dots, O_m$ = the set of possible operations that can be included in the robotic system
- $W_1, \dots, W_n$ = the set of workstations that can be purchased to process the operations
- c_j = the purchase cost of workstation W_j
- B = the capital investment budget
- $O(j)$ = the index set of operations that can be performed by workstation W_j
- $W(i)$ = the index set of workstations that can perform operation O_i
- t_{ij} = the time for workstation W_j to perform operation O_i
- h_j = the number of available hours at workstation W_j within time horizon T , allowing for maintenance and other unavoidable downtime
- r_i = the number of times operation O_i is required within time horizon T
- v_{ij} = the net value, i.e., revenue less cost, produced by operation O_i at workstation W_j

We also define the following decision variables:

$$Y_j = \begin{cases} 1 & \text{if workstation } W_j \text{ is purchased} \\ 0 & \text{otherwise} \end{cases}$$

$$X_{ij} = \text{the number of times operation } O_i \text{ is performed at workstation } W_j \text{ within time horizon } T$$

With these data and variable definitions, the preliminary design problem of maximizing the value of operations performed in the robotic system can be written as follows.

$$\text{maximize } \sum_{i=1}^m \sum_{j \in W(i)} v_{ij} X_{ij} \quad (1)$$

$$\text{subject to } \sum_{j=1}^n c_j Y_j \leq B \quad (2)$$

$$\sum_{j \in W(i)} X_{ij} \leq r_i, \quad i = 1, \dots, m \quad (3)$$

$$\sum_{i \in O(j)} t_{ij} X_{ij} \leq h_j Y_j, \quad j = 1, \dots, n \quad (4)$$

$$X_{ij} \geq 0 \quad \text{and integer,} \quad i = 1, \dots, m, \quad j = 1, \dots, n \quad (5)$$

$$Y_j \in \{0, 1\}, \quad j = 1, \dots, n \quad (6)$$

The objective (1) represents the total value produced. It sums the value produced by operation O_i on workstation W_j over all operations and workstations. Constraint (2) ensures that the total capital investment does not exceed the budget, B . Constraints (3) ensure that the number of times an operation is performed does not exceed its production requirement. Constraints (4) ensure that the total workload assigned to workstation W_j does not exceed the available hours there, which are h_j if W_j is purchased and 0 otherwise. Constraints (5) ensure that the number of times operation O_i is assigned to workstation W_j is a nonnegative integer. Finally, constraints (6) enforce the choice either to purchase workstation W_j or not to do so. This model is useful in that it can evaluate, for example, the tradeoff between expensive workstations that are capable of performing many operations and cheaper but less flexible workstations.

The above integer program may be difficult to solve for large data sets, either by using standard software or with a customized technique. The reason is that the computational difficulty involved in solving this model increases very rapidly with the number of integer variables. The reader may refer to the introductory chapters of Garey and Johnson (1979) for a more detailed explanation. Note that the above formulation has $mn + n$ integer variables.

To circumvent this computational difficulty the following approximation is often useful. We remove the integer requirement from Constraints (5), thus requiring only that $X_{ij} \geq 0$. This greatly reduces the number of integer variables to n , and consequently also the difficulty of solution. For example, if $m = n = 30$, solving the original integer program with 930 variables is a major challenge that requires expensive software and probably a long running time on a mainframe computer, whereas the new *mixed integer* program has only 30 integer variables and can easily be solved on a personal computer. Moreover, it is frequently the case that few X_{ij} variables in the mixed integer program have fractional values (e.g., $X_{11} = 2.4$), and that these can be rounded to integer values (e.g., $X_{11} = 2$ or 3) without serious detriment to the quality of the overall solution.

Applications of integer programming to flexible manufacturing systems are discussed by Stecke (1983) and Berrada and Stecke (1986). Commercially available software packages for integer and mixed integer programming are reviewed by Saltzman (1994).

2.3.2 Linear Programming

Perhaps the most important contribution made by operations research to decision-making is the technique of *linear programming*. Bradley, Hax, and Magnanti (1977), Schrijver (1986), and Bazaraa, Jarvis, and Sherali (1990) provide very detailed treatments of the theoretical background of linear programming. As an example of the power of this technique, Bixby et al. (1992) describe the solution of an extremely large problem with 12,753,313 decision variables and 837 constraints. Some exciting recent developments in this area are reviewed and evaluated by Lustig, Marsten, and Shanno (1994). For a very readable discussion about how to build practical models that access the power of this approach, the reader may refer to Williams (1990).

Because of the existence of highly efficient solution methods, which can be implemented in inexpensive commercial software, linear programs with many thousands of variables and constraints are readily solvable. Linear programming offers less modeling flexibility than mixed integer programming, in that all the variables are considered divisible. However, there are many applications where this limitation is not very problematic.

To illustrate the use of linear programming, we consider a subproblem of the problem from Section 2.3.1. In this subproblem the choice of workstations has already been made. It only remains to decide how to allocate the available operations to those workstations.

Using the notation of Section 2.3.1, the value maximization problem can be written as follows: maximize (1), subject to (3), (4') and (5'). Constraints (4') are formed from Constraints (4) by setting $Y_j = 1$ if workstation j was purchased, and $X_{ij} = 0$ for $i = 1, \dots, m$ if workstation j was not purchased. Constraints (5') are formed from Constraints (5) by removing the integer requirement on the X_{ij} variables, thus requiring only that $X_{ij} \geq 0$. Note that the integer Y_j variables and all the constraints in which they appear are no longer present. Thus the problem can be solved as a linear program, with much less time and cost than the formulation in Section 2.3.1, subject to the same issues about fractional X_{ij} values as were discussed there.

Closely related to linear programming is *goal programming* (Charnes, Cooper, and Ferguson, 1955; Ignizio, 1976), which allows the modeling of conflicting goals or objectives. Imany and Schlesinger (1989) demonstrate the use of goal programming to optimize the robot selection process taking into account requirement priorities.

Schrage (1991) discusses one very popular linear programming software package in detail. Fourer (1997) presents a comprehensive review of commercially available software packages for linear programming. A valuable source of on-line information about linear programming, including free software and discussions about solvability, is the Frequently Asked Questions Homepage maintained by J. Gregory at "<http://www.mcs.anl.gov/home/otc/Guide/faq>".

2.3.3 Analytic Hierarchy Process

The *analytic hierarchy process*, developed by Saaty (1980), is a technique for making comparisons between several options across multiple factors. The difficulty in such comparisons arises when the relative performance of the options is inconsistent across the various factors. Since the number of robotic systems being compared is likely to be quite small, and since that comparison involves several factors such as cost, space utilization, and flexibility (see Section 2.1), the analytic hierarchy process is ideally suited to help with the choice of which system to purchase. Saaty (1986) presents an in-depth discussion of the theoretical foundations of the analytic hierarchy process and addresses some concerns of earlier authors about its correctness. Zahedi (1986) and Golden, Wasil, and Levy (1989) give bibliographies of its applications.

As an illustration, let us assume that we wish to compare three robotic systems (A, B, and C) using the three factors mentioned above. This creates a decision-making hierarchy in which the choice of which system to purchase is on the top level, the three factors mentioned are on the second level, and the individual systems are on the third level. This hierarchy is illustrated in Figure 1.

The analytic hierarchy process is implemented as follows. The decision-maker provides various *pairwise comparisons* between factors, indicating which is more important and how much more, using a discrete scale. Also, within each factor similar pairwise comparisons are made regarding how well each system performs for that factor. The outcome of the analysis is a useful overall ranking of Systems A, B, and C. This technique has the advantage that responses on a pairwise comparison point scale are often more reliable than those on an abstract interval scale. A further advantage is that the technique identifies inconsistent input data, thereby giving the decision-maker a chance to review and correct

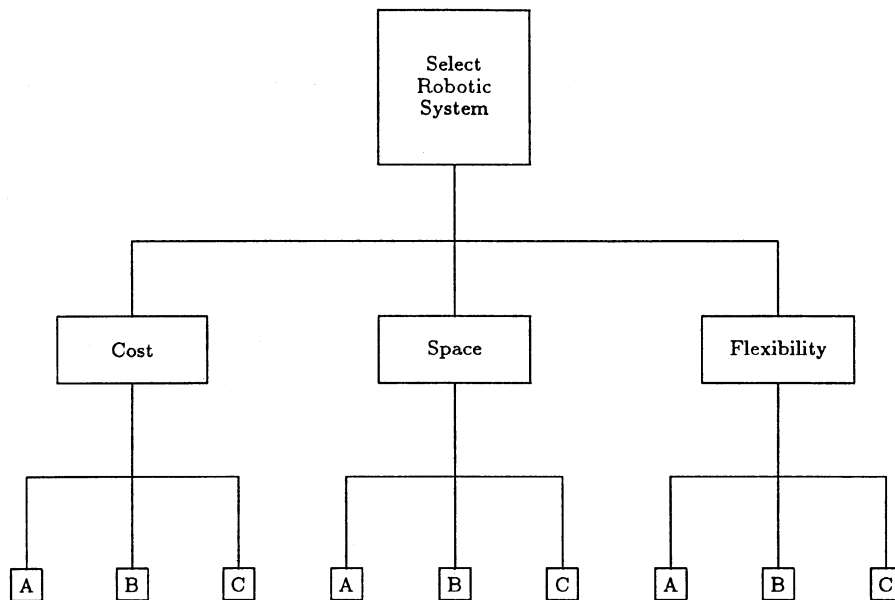


Figure 1 Robotic system choice using the analytic hierarchy process.

it. The analytic hierarchy process is less suitable than mixed integer or linear programming for making comparisons between thousands of alternatives that differ only slightly, or for dealing with constraints, such as might be necessary in detailed production planning. However, it can be a valuable aid to comparing several options across several factors.

Applications of the analytic hierarchy process to manufacturing systems are described by Arbel and Seidmann (1984), Varney, Sullivan, and Cochran (1985), Bard (1986), and Sullivan (1986). Seidmann, Arbel, and Shapira (1984) use the analytic hierarchy process to assist in robotic system design. Frazelle (1985) presents an application to choices among material-handling alternatives. Shang and Sueyoshi (1995) use data envelopment analysis (Farrell, 1957; Charnes, Cooper, and Rhodes, 1978; Sengupta, 1995) to unify an analytic hierarchy process model and a simulation model (see Section 3.3.4) for the problem of selecting a flexible manufacturing system. An interesting new variant of the analytic hierarchy process is described by Lipovetsky (1996). A popular analytic hierarchy process software package is EXPERT CHOICE.

2.3.4 Heuristics

Models of the type described in Section 2.3.1 may be computationally hard to solve to optimality, as discussed by Garey and Johnson (1979). Since optimality is hard to achieve, a useful “second-best” approach is to solve those problems *heuristically*. A heuristic is a simple problem-solving approach that may not give an optimal solution. Many computer codes for integer and mixed integer programming problems allow the user to terminate the search process when it becomes very lengthy and to identify the best solution found so far. Typically the heuristic solution found in this way will be within a few percentage points of optimal, which for many practical purposes may be good enough. Examples of heuristic design and evaluation are provided by Talbot and Gehrlein (1986) and Adams, Balas, and Zawack (1988). The theoretical state of the art in this area is comprehensively examined in Hochbaum (1997).

It is also possible to design heuristics directly for decision problems. As an example, consider the problem of choosing which of n workstations to buy. Here the net present value (respectively, capital cost) of workstation W_j is denoted by b_j (respectively, c_j). The total capital cost must not exceed a given budget, B . We can model the problem of maximizing the total net present value using the following integer program.

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n b_j Y_j \\ &\text{subject to} && \sum_{j=1}^n c_j Y_j \leq B \\ &&& Y_j \in \{0, 1\}, \quad j = 1, \dots, n \end{aligned}$$

In this model $Y_j = 1$ if workstation W_j is purchased, and $Y_j = 0$ otherwise. This model is known as a 0–1 knapsack problem (Balas, 1980), which may be hard to solve computationally if n is large. Moreover, additional constraints or complications may arise in practical decision problems involving robotic facilities. Examples include space restrictions, material flow capacities, or the opportunity for discounts with multiple purchases. However, the following well-known heuristic often provides solutions that are close to optimal for the above formulation.

Greedy Heuristic

- Step 1.** Order the workstations such that $b_1/c_1 \geq \dots \geq b_n/c_n$.
- Step 2.** Consider the next workstation in index order. If enough of the budget remains to purchase it, then do so.
- Step 3.** If not all workstations have been considered, then go to Step 2. Otherwise, stop.

The Greedy Heuristic is intuitive in that it first considers workstations with larger ratios of net present value to capital cost, that is, the most attractive workstations.

Hochbaum and Maass (1984) describe a variety of heuristic approaches to robotics and related problems. Heuristics for layout problems in automated manufacturing are provided by Heragu and Kusiak (1988) and Kouvelis and Kim (1992). Ball and Magazine

(1988), Ahmadi and Tang (1991), Lofgren, McGinnis, and Tovey (1991), and Lane and Sidney (1993) describe and evaluate heuristics for automated manufacturing systems. Sethi et al. (1990) discuss heuristics for the selection and ordering of part-orienting devices in flexible manufacturing systems.

2.3.5 Location and Layout

The main *location* decision is where to build a new robotic facility. The following factors are particularly important in such decisions:

1. Transport costs from/to other facilities
2. Fixed costs
3. Production capacity, as determined by site characteristics

A typical facility location problem is illustrated in Figure 2. The objective is to locate a single new robotic facility as close as possible to three markets and three suppliers. However, only sites A, B, C, and D are available, and the choice between them involves a comparison of costs and travel distances and times.

The operations research literature includes a large number of studies on location problems. An overview is provided by Francis, McGinnis, and White (1983). Schmenner (1982) discusses how large companies make location decisions and concludes that key factors are labor costs, unionization, closeness to markets and suppliers and other company facilities, and quality of life.

An important choice in modeling location problems is how to measure distance, using either a straight-line or a rectilinear measure. The latter measure is based on the assumption that travel is only possible in the north-south or east-west directions, as might be the case in urban planning problems. Assuming that location at any point on a plane is possible typically makes location problems easier to solve, as for example in a marketing application by Gavish, Horsky, and Srikanth (1983). However, for many applications, only a discrete set of possible locations is available. The intermediate and broadly applicable case of location on networks is discussed by Handler and Mirchandani (1979). Among the best known studies of location problems are those of Erlenkotter (1978) and Ross and Soland (1978). Jungthirapanich and Benjamin (1995) develop a knowledge-based decision support system for facility location. The issues arising in the location of automated facilities are not very different from those arising in facility location generally, and therefore the reader should find much of the literature on this topic to be directly useful.

The main *layout* decisions are where, within an existing plant, to locate a robot-served workstation and its peripheral equipment such as storage devices. In this context the layout

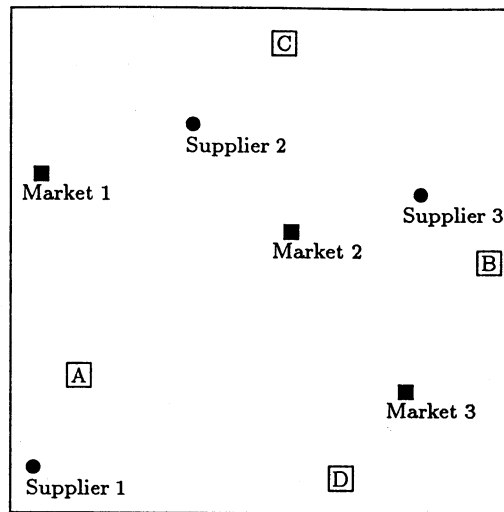


Figure 2 Robotic facility location problem.

problem is very similar to the location problem shown in Figure 2. Proximity to other workstations is in general desirable, but space and material-handling limitations may limit the available options. Relevant factors include:

1. Transport time from/to other workstations within the plant
2. Dimensions of the workstation
3. Limited reach of transporters, e.g., robot arms
4. Production capacity, as determined by material flow bottlenecks

Francis, McGinnis, and White (1992) provide a comprehensive discussion of both location and layout problems. They describe several basic horizontal flow patterns (for use within a single floor) as well as vertical flow patterns (for use across several floors). Layout analysis (Apple, 1977; Tompkins et al., 1996; Heragu, 1997) often begins by establishing a relationship chart that indicates the desirability (or occasionally, undesirability) of locating two operations close to one another. A from-to chart is used to identify the trip frequencies between workstations. These two charts can be used as input information for a variety of techniques, many of which are implemented in commercial software. Among the better known packages are CRAFT 3-D (Cinar, 1975), SPACECRAFT (Johnson, 1982), BLOCPLAN (Donaghey and Pire, 1991) and MULTIPLE (Bozer, Meller, and Erlebacher, 1994). Francis et al. (1994), extending earlier work by Drezner and Nof (1984), consider the problem of finding the component bin locations and part sequence that will minimize total time in a robotic assembly problem. Genetic search (Michalewicz, 1996) is a heuristic search procedure based on an analogy with a natural evolutionary process in which intermediate solutions are allowed to “reproduce” new and hopefully improved solutions. Tate and Smith (1995) use genetic search techniques to develop a facility layout model. Dynamic layout problems, in which changes in the data motivate changes in the layout over time, are considered by Rosenblatt (1986) and Urban (1993). Neural networks (Aleksander and Morton, 1990; Warwick, 1995) are also based on a biological analogy, and perform searches by adaptive parallel processing of information. Tsuchiya, Bharitkar, and Takefuji (1996) present and test a heuristic neural network procedure for facility layout. A design approach that features parallel and continual interaction between product, process and system design is recommended by Rampersad (1995) for a robotic assembly system layout problem. An interesting survey that highlights research trends and gives an overview of several useful solution procedures for layout problems is provided by Meller and Gau (1996).

Scriabin and Vergin (1975) sparked an interesting debate about the relative effectiveness of human layout planners and computer packages. They conclude that, in large problems, the superior pattern recognition abilities of a human planner lead to more effective layouts. However, further studies (Block, 1977) and subsequent developments in artificial intelligence (Brown and White, 1990) argue against such a conclusion.

3 SYSTEM DESIGN

In this section we assume that, as a result of decisions made in the system planning phase, we have already identified a small number of options for the set of operations to be automated and the set of robot-served workstations to be purchased. Whereas the data used at the system planning stage may have been aggregate and approximate, in the system design phase both more detail and more accuracy are required. In particular, the characteristics of individual operations need to be considered so that the related material handling requirements can be met within the system that is eventually adopted.

3.1 Issues and Requirements

The main issues and management requirements that need to be considered at the system design phase are as follows.

1. *Equipment layout.* A good layout should be found for automated workstations, storage facilities, and personnel, taking into account patterns of material flow within the robotic system and within other systems that interface with it. Related operations research techniques are discussed in Section 2.3.5.
2. *Cell design.* If a cellular layout is to be used, a decision is needed about whether cells should be organized by product or by process. Procedures for forming groups of operations should be developed.

3. *Material-handling equipment.* A decision is needed about what material-handling equipment should be purchased. A similar decision is needed about what storage equipment, such as an automated storage and retrieval system, to be purchased.
4. *Accessories.* It is necessary to decide the number and types of peripheral equipment, including part feeders and part orienters, magazines, and tool holders, to be purchased.

3.2 Performance Measures

The performance measures that are useful at the system design phase fall into two types. First, they include refinements of the measures suggested for the system planning phase. Second, they include new measures that can now be considered because of the greater detail in the available data.

1. *Throughput rate.* For each cell or for the overall system a natural objective is to maximize the throughput rate. In modern manufacturing systems it is common to use a *minimal part set*, which is the smallest possible set of parts having the same proportions as the overall production target. Minimizing the cycle time to produce a minimal part set repetitively is equivalent to maximizing the throughput rate.
2. *Turnaround time.* This is the elapsed time from the entry of a job into the robotic system until it leaves, and is a measure of system responsiveness.
3. *Work-in-process inventory.* Since holding cost may be a significant part of overall manufacturing cost, there is a need to control work-in-process inventory. This relates closely to buffer storage design also. Further, a result to be discussed in Section 3.3.2 suggests that in a steady state the average turnaround time is proportional to the average work-in-process inventory.
4. *Equipment utilization.* A traditional measure of manufacturing effectiveness is the minimization of equipment idle time. A more modern view, however, suggests that this is less useful than the other measures above.
5. *System reliability.* It is important to understand how the reliability of the different components of the robotic system will interact in various configurations, and thus to assess the reliability of the overall system.

3.3 Operations Research Techniques for System Design

Because of the availability of more accurate data, the operations research techniques discussed here are capable of greater accuracy than those used at the system planning phase. However, estimates may still be needed for the requirements of individual operations, for example. Therefore, it is important to find ways to perform sensitivity analysis or investigate “what if” scenarios for the information which is output by those techniques.

3.3.1 Group Technology and Cell Formation

In *group technology* (Mitrofanov, 1966; Burbidge, 1975) a major decision is how to decompose the manufacturing system into *cells*. Families of component parts are identified and each family is fully or predominantly processed within a machine group. The most obvious advantage of group technology is setup time reduction arising from the similarity of the parts within the same cell. Other advantages include improved productivity and quality, as well as reduced inventories and tool requirements. In the remainder of this subsection we summarize the main approaches to cell formation. Singh (1993) provides a useful review of the literature on this topic.

Coding and classification schemes work by coding parts according to their attributes. Bedworth, Henderson, and Wolfe (1991) provide a summary of such schemes. However, there is relatively little work that has integrated such schemes with more formal operations research techniques. Kusiak (1985) uses a model from location theory to achieve this integration.

In machine-component grouping analysis methods, cells are formed by algebraic operations on a 0–1 machine-component index. Chu and Pan (1988) provide a review of these methods, which lack the ability to deal effectively with bottleneck machines, and do not consider operating details such as processing times. Ng (1996) shows how to separate machines and components into independent families when such an ideal separation is possible, and then uses this information to design a cell formation procedure that performs very well on standard test problems.

Similarity-based clustering techniques design measures of similarity and then cluster jobs in various ways. For example, Rajagopalan and Batra (1975) cluster jobs using graph theory. Other mathematical approaches that eschew the use of similarity coefficients include a decomposition scheme and heuristics (Askin and Chiu, 1990) and a network approach (Vohra et al., 1990). These approaches often suffer from the problem with bottleneck machines mentioned in the previous paragraph.

Other potentially valuable techniques include a knowledge-based system (Kusiak, 1988) and a neural network approach (Kaparthi, Suresh, and Cervený, 1993). Burke and Kamal (1995) demonstrate the use of the fuzzy adaptive resonance theory neural network. Xie (1993) focuses on the minimization of intercell traffic, a problem for which three heuristics are designed and tested. Methods for reconfiguring existing cells are discussed by Mohamed (1996).

3.3.2 Queueing Models

As mentioned in Section 3.2, an important performance measure for robotic systems is work-in-process inventory. This varies constantly while the system is operating, due to the random processes of jobs arriving and jobs being completed. However, *queueing models* can take this random behavior into account and provide useful performance measures. When used in this way, a queueing model is a probabilistic, steady-state model. Jobs waiting to be processed are essentially “customers” waiting in line while other jobs are being processed. General introductions to queueing models are provided by Kleinrock (1975, 1976) and Gross and Harris (1985).

Queueing models can help in evaluating how much service should be provided. Within the context of robotic systems, this means how many workstations and with what capabilities (e.g., material-handling capacities and processing speeds) and how many operators. A service level that is too high incurs unnecessary capital and operating costs, but one that is too low creates expensive work-in-process inventories. Queueing models do not optimize this design question directly. However, given a particular system configuration, they can provide several steady-state measures of its performance.

In order to present some useful results for queueing models, we define the following notation:

- ρ = utilization factor of the system
- λ = average arrival rate of the jobs
- L_q = average number of jobs waiting to be processed
- L = average number of jobs waiting or in process
- W = average total of waiting and processing time of a job
- μ = average service rate per server
- m = number of servers

The utilization factor is defined to be the average arrival rate divided by the total average service rate, or $\rho = \lambda/m\mu$. A queueing system is described using the three field notation $A|S|m$, where A describes the arrival process, S describes the service process, and m is the number of servers. Some common types of arrival or service process appearing in the first two fields include: M = Markovian (times are independent and exponentially distributed), G = general (times follow an arbitrary distribution), and D = deterministic (all times are the same).

The most important result in queueing models, initially proved by Little (1961) for a simple system and then extended by Stidham (1974) to more general systems, is:

$$L = \lambda W \quad (7)$$

In the context of robotic systems, Equation (7) implies that the work-in-process inventory is proportional to the average amount of time a job spends in the system. This relationship emphasizes the importance of timely throughput of work-in-process.

Single-Robot System

We can consider a single-robot system as a queueing model with only one server. Figure 3 illustrates a system of this type. Jobs arrive from other workstations independently and according to an exponential distribution. For a general service time distribution this system is denoted $M|G|1$.

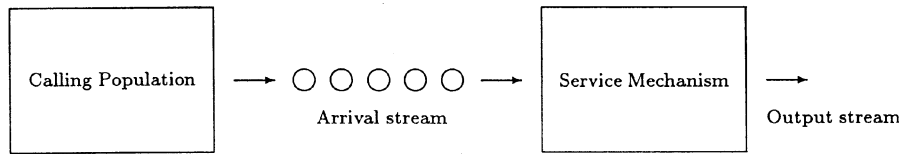


Figure 3 Single-server queueing system.

The following formula gives the average number of jobs waiting in an $M|G|1$ system (Kleinrock, 1975):

$$L_q = \rho^2(1 + C_s^2)/[2(1 - \rho)]$$

where C_s denotes the coefficient of variation (i.e., the standard deviation divided by the mean) of the service times. However, a common occurrence in robotic systems is that all jobs require the same operation by the robot. In this case we have an $M|D|1$ queueing model where $C_s = 0$, and thus

$$L_q = \rho^2/[2(1 - \rho)] \quad (8)$$

It is important to note that Equation (8) is nonlinear. In fact, if ρ starts to increase from a low level such as 0.1, L_q increases slowly at first but then more and more rapidly. Table 2 shows the values of L_q associated with various values of ρ in an $M|D|1$ system.

If the jobs arrive from other workstations, then an approximation that is usually satisfactory is that both the times between job arrivals and the service times are exponentially distributed. This situation is known as an $M|M|1$ system, for which it can be shown (Trivedi, 1982) that

$$L_q = \rho^2/(1 - \rho) \quad (9)$$

Multiple Robot System

The results from the single robot system can be extended to a system with m identical processors, where each job can be performed by any processor. A system of this type is illustrated in Figure 4. Following the above assumption about exponentially distributed interarrival and service times gives an $M|M|m$ system, for which it can be shown (Trivedi, 1982) that

$$L_q = \rho(mp)^m P_0 / [(1 - \rho)^2 m!] \quad (10)$$

where P_0 is the long-run average proportion of time for which all m servers are idle, and where P_0 has a closed form expression.

Pooling of Facilities

A very interesting issue in queueing models is whether to pool facilities. Suppose we wish to process two types of jobs using two robot-served workstations. A choice between two alternative designs presents itself. In the first design one workstation is dedicated to each type of job. In the second design the workstations are pooled and each is used for both types of jobs. The first (dedicated workstation) design is likely to be less expensive in capital cost. This is because the second design may need duplicated equipment such as part orienters and specialized gripping devices to handle both job types. However, it can be shown from Equations (9) and (10) that the first design has more jobs waiting on average. For example, if $\rho = .75$ and the two types of jobs arrive at the same rate, there

Table 2 Average Number of Jobs Waiting in an $M|D|1$ System

ρ	.15	.25	.35	.45	.55	.65	.75	.85	.95
L_q	.0132	.0417	.0942	.1841	.3361	.6036	1.125	2.4083	9.025

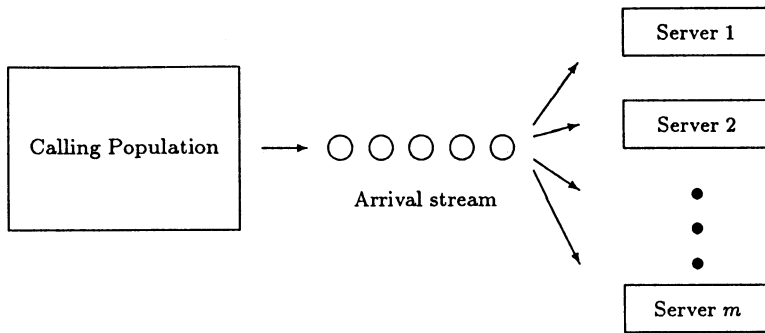


Figure 4 Multiple-server queueing system.

are on average 2.25 extra jobs (4.5 vs. 2.25) waiting in the first design, creating an increase of 100% in work-in-process.

Despite this significant mathematical advantage of a combined service system, Rothkopf and Rech (1987) make a persuasive behavioral case for the first design above in queueing systems where the customers are human, rather than jobs. Some of their arguments, moreover, such as the relative proximity of the waiting and service areas in the first design, can also be applied to robotic systems.

Ram and Viswanadham (1990) and Viswanadham and Narahari (1992) discuss the use of queueing models to evaluate the performance of a flexible machine center problem.

3.3.3 Queueing Networks

Queueing networks extend the methodology of queueing models discussed in Section 3.3.2. In doing so they provide a much greater ability to model the complexities of real automated manufacturing systems. Jobs that in general may have different routings through the available workstations are transported, by a robot or other automated material handler, from one workstation to another. Comprehensive introductions to queueing networks are provided by Walrand (1988), Viswanadham and Narahari (1992), Buzacott and Shanthikumar (1993), and Gershwin (1994). Extensive surveys are provided by Bitran and Dasu (1992) and Buzacott and Shanthikumar (1992).

There are two basic types of queueing networks: open and closed. An open queueing network, illustrated in Figure 5, has one or more points at which customers enter the system and one or more points at which they leave the system after processing. By contrast, in a closed queueing network, as illustrated in Figure 6, customers neither enter nor leave the system. In a manufacturing context such customers might represent parts in an open network. In a closed network the customers might represent pallets that accompany parts through the network. Little's (1961) result in Equation (7) extends to queueing networks and implies that the fraction of waiting time at any workstation is equal to the fraction of work-in-process at that workstation.

Jackson (1957, 1963) gives an important result about open queueing networks in which an arriving job enters the network at workstation i (which has m_i processors) with probability q_i , for $i = 1, \dots, m$. After processing, the job is routed to workstation j with probability p_{ij} . Let $P(k_1, \dots, k_m)$ denote the steady-state probability that there are k_i jobs at workstation i for $i = 1, \dots, m$, and let $P_i(k_i)$ denote the probability that there are k_i jobs in a one workstation $M/M/1$ queueing system. Then Jackson's theorem states that

$$P(k_1, \dots, k_m) = \prod_{i=1}^m P_i(k_i)$$

The importance of this result is that it enables the use of standard queueing theory (Section 3.3.2) to derive performance measures for queueing networks.

Analogous results for closed queueing networks are due to Gordon and Newell (1967). Both types of model are known as *product form* queueing networks. This class was extended and presented in a unified way by Baskett et al. (1975), and within the extended

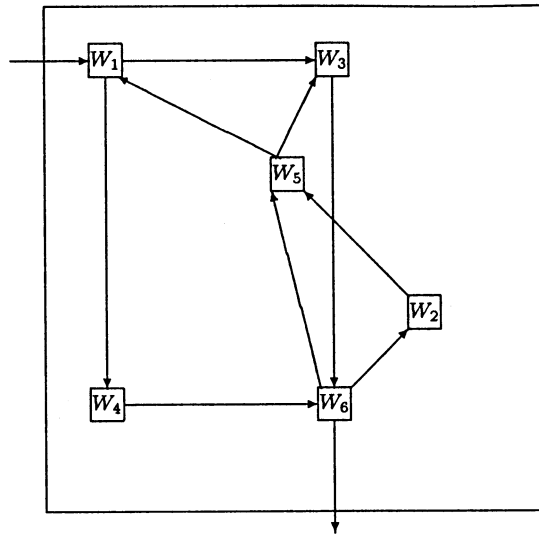


Figure 5 Open queueing network.

class many measures of system performance can easily be obtained. However, a number of features of automated manufacturing systems may violate the product form requirement and thus require the use of approximations. These features include nonexponential service times, nonstandard customer service orders, production deadlines, assembly product structures, breakdowns, the use of dynamic rerouting of jobs to avoid bottlenecks, and resource constraints.

The following types of queueing networks have been studied extensively for both open and closed cases: tandem queues (where machines are in series), networks with feedback (such as items returned for rework after inspection), networks with blocking (where a job waits at a workstation until the next workstation is ready), open central server networks, and product form networks. Additionally, product form networks allow the analysis of problems with multiple job classes.

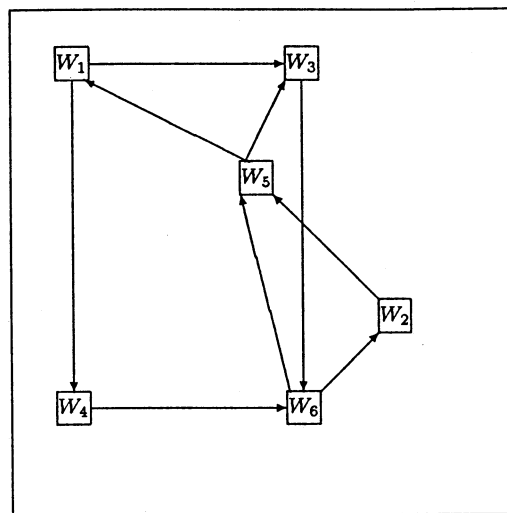


Figure 6 Closed queueing network.

Solberg (1977), Suri and Hildebrandt (1984), and Yao and Buzacott (1987) provide examples of the application of queueing networks to automated manufacturing systems. Chen et al. (1988) demonstrate the accuracy of performance predictions from a queueing network model of a semiconductor wafer fabrication facility. A survey of sensitivity analysis results is provided by Liu and Nain (1989). Bitran and Tirupati (1989) analyze tradeoff relationships between different performance measures in manufacturing queueing networks. Dallery and Stecke (1990) and Calabrese (1992) address the issue of how to allocate workloads and servers in similar environments. Software packages that are available to analyze queueing networks include CAN-Q (Solberg, 1980) and MANUPLAN (Suri and Diehl, 1985). Seidmann, Schweitzer, and Shalev-Oren (1987) perform a comparative evaluation of four queueing network software packages. A variety of methods for analyzing queueing networks are described by Robertazzi (1994).

3.3.4 Simulation

Simulation is a controlled statistical sampling technique for evaluating the performance of complex probabilistic manufacturing and other systems. General introductions to simulation techniques can be found in Law (1986), Law and Kelton (1991), and Schriber (1991).

With simulation, the performance of a real system is *imitated* by a computer model that uses preestablished probability distributions to generate random events (such as demand rates, processing times, and machine breakdowns) that affect the performance of the system. The outcome of a simulation study is statistical information about the performance of the system. When applied to large manufacturing systems, simulation tends to generate and process a huge amount of data, which can be time-consuming and expensive compared to the use of most other operations research techniques.

In order to use simulation to assess the performance of a robotic system, it is first necessary to collect information about the various hardware and software components of the system. The information to be collected includes both deterministic information (such as known processing times for standard parts) and probabilistic information (such as demand rates for new products). For the probabilistic information it is necessary to fit the data collected to a probability distribution using standard statistical techniques (Kleijnen, 1987). These distributions are used to generate specific events, the effects of which are simulated by a computer model that captures as many of the intricacies of the real operating system as possible. The tradeoff between realism and solvability is less problematic in simulation than in the optimization approaches discussed in Sections 2.3.1 and 2.3.2 because simulation possesses much greater flexibility and can therefore be extended, without significant loss of solvability, to allow for complex realities.

There are essentially three approaches to simulating a robotic system:

1. *Simulation package.* This approach (Runner and Leimkuhler, 1978; Phillips and Handwerker, 1979) involves the purchase of a manufacturing system simulation package, which can then be modified to approximate the real system more accurately. While this approach is likely to be relatively quick and simple, it may not be possible to introduce all the complexities of a specific real-world system into a particular package.
2. *Simulation language.* This relatively sophisticated approach (Gordon, 1975; Pritsker and Pegden, 1979) requires knowledge of a simulation language. Also, there may be interface problems with system subroutines that are written in a standard programming language such as Fortran. The main advantage of this approach, however, is that it offers almost unlimited modeling flexibility. Thus, a real-world system can be modeled as accurately as desired.
3. *Standard programming language.* This approach takes advantage of already existing knowledge of a standard programming language such as Pascal, which is used to write a simulation code for the system. Interfacing with system subroutines is easy and the simulation package is universally portable. The modeling flexibility of this approach is similar to that of using a simulation language. However, all the basic components must be written by the user, and the overall programming task is likely to be much greater.

Further discussion of these tradeoffs appears in Bevans (1982). The use of simulation to evaluate the performance of manufacturing systems is discussed by Halder and Banks

(1986), Carrie (1988), and Trunk (1989). An example of the use of simulation to plan cellular layouts is provided by Morris and Tersine (1990). De Lurgio, Foster, and Dickerson (1997) use simulation to assist in the selection of equipment for integrated circuit manufacturing. A useful survey of simulation practice in industry is provided by Christy and Watson (1993). Hlupic and Paul (1995) conduct a thorough case study evaluation of four manufacturing system simulation packages: WITNESS, EXCELL+, SIMFACTORY II.5, and ProModel PC. None of the packages dominates the others with respect to all performance criteria, but they discuss the relative advantages of each. Kleijnen (1995) and Kleijnen and Rubinstein (1996) discuss methods for validating a simulation analysis and performing sensitivity analysis on the results. A comprehensive survey of commercially available simulation software packages appears in Swain (1995). Banks (1996) provides useful advice about how to interpret checklists of features when buying a simulation software package. A survey of simulation packages, including some that are designed specifically for robotic systems, appears in the May 1997 issue of *IIE Solutions*. Application-specific packages, such as IGRIP and TELEGRIP, may capture more details of robotic systems, or may require less programming to do so, than general-purpose manufacturing simulation packages.

3.3.5 Reliability and Maintenance

The *reliability* of a robotic system is critical because the substantial investment in capital cost needs to be repaid through consistent output of high-quality products. Thus, in evaluating a robotic system, it is important to be able to estimate how frequently and for how long it will be unavailable due to component failures. Methods for making such estimates fall within the topic of reliability theory. Classic introductory texts in this area are provided by Barlow and Proschan (1965, 1975).

Single-Component Systems

In order to illustrate the use of reliability theory, we define the following notation. Let T be a random variable that represents the lifetime of a component. Let $F(t)$ denote the cumulative distribution function of T , i.e., $F(t) = P\{T \leq t\}$. Assuming that T is differentiable, let $f(t)$ denote its density function, i.e., $f(t) = dF(t)/dt$. A valuable concept is the failure rate function $r(t)$, which is defined by

$$r(t) = f(t)/[1 - F(t)]$$

For small values of Δ the probability that a component which has survived to time t fails by time $t + \Delta$ is given by $r(t)\Delta$. The failure rate function can be increasing, constant, or decreasing over time.

It is also possible to derive closed form expressions for the mean T_a and the variance V_a of the time for which the system is available during a time horizon of length T . Let f (respectively, s_f) denote the mean (respectively, standard deviation) of the time from the last repair to the next failure. Let r (respectively, s_r) denote the mean (respectively, standard deviation) of the time from the last failure to the next repair. Then we have

$$T_a = fT/(f + r) \quad (11)$$

$$\text{and } V_a = [(fs_f)^2 + (rs_r)^2]/(f + r)^3$$

Equation (11) is highly intuitive, in that the long-run proportion of time in which the system is available is $f/(f + r)$, i.e., the mean time from a repair to the next failure divided by the mean time between two successive repairs.

Multicomponent Systems

Reliability theory also helps us to evaluate the performance of systems with multiple components. Suppose that there are m components, where R_i is defined as the proportion of time in which the i th component is available. Note that R_i represents the reliability of the i th component. The following analysis assumes independence of the availability of the components. This assumption greatly simplifies the mathematics involved in estimating the reliability of the system, which might otherwise defy exact solution procedures. It should be pointed out that the independence assumption is critical and, if made erroneously, can lead to poor predictions. For example, Nahmias (1993) identifies this as a

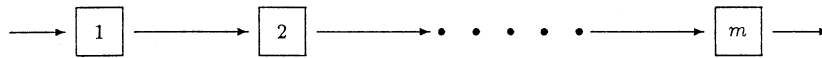


Figure 7 Series structure.

source of error in a U.S. Nuclear Regulatory Commission (1975) study of reactor safety prior to the 1979 Three Mile Island accident.

Three possible designs for multicomponent systems are (a) a series structure, which is the least reliable, (b) a parallel structure, which is the most reliable, and (c) a hybrid structure.

A series structure is illustrated in Figure 7. Here the system fails unless all components are available. The reliability of the system, R_s , is thus given by

$$R_s = \prod_{i=1}^m R_i$$

For example, if $m = 6$ and $R_1 = \dots = R_6 = .9$, then $R_s = 0.531441$, which is much too low for most applications.

A parallel structure is illustrated in Figure 8. Here the system fails only if all components fail. Thus, the reliability of the system is given by

$$R_s = 1 - \prod_{i=1}^m (1 - R_i)$$

In the above example we obtain $R_s = 0.999999$, a very high level of reliability.

One example of a hybrid structure is illustrated in Figure 9. This system fails unless at least one out of each pair of components (1, 2), (3, 4), $\dots$, $(m-1, m)$ is available. The reliability of this structure is given by

$$R_s = \prod_{i=1}^{m/2} [1 - (1 - R_{2i-1})(1 - R_{2i})]$$

In the above example we obtain $R_s = 0.970299$.

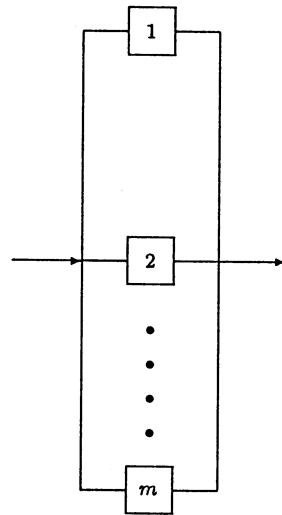


Figure 8 Parallel structure.

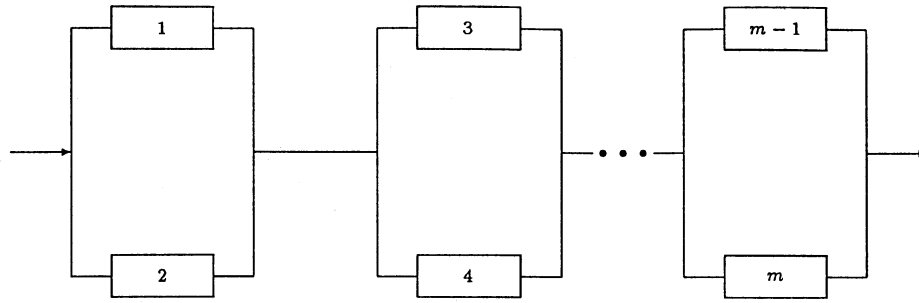


Figure 9 Hybrid structure.

Another common type of hybrid structure fails if fewer than any k out of the m components are available. Here, assuming that $R_1 = \dots = R_m = R$, the reliability of the system is given by

$$R_s = \sum_{j=k}^m m! R^j (1-R)^{m-j} / j!(m-j)!$$

In the above example for the case $k = 3$, we obtain $R_s = 0.99873$.

It should be noted that the reliability of both hybrid structures is close to that of the parallel structure. However, because of the multiple ways in which a hybrid structure can be designed, it is often more flexible and cheaper than a parallel structure.

Prasad, Nair, and Aneja (1991) discuss how to assign components in multicomponent systems to maximize reliability. Hwang and Rothblum (1994) consider the problem of assigning interchangeable parts of many types to modules with a series structure and prove that the best parts of each type should be assigned to the same module. Chung and Flynn (1995) discuss how to identify optimal replacement policies in hybrid structures.

Maintenance and Planned Replacement

A survey of *maintenance* models is provided by Pierskalla and Voelker (1976). Vanneste and Van Wassenhove (1995) review recent developments in maintenance theory and practice, and propose an integrated approach that combines some of the best features of other methods. Feldman and Chen (1996) perform some postoptimality analyses for various policies. A computational comparison of different maintenance policies is performed by Kelly, Mosier, and Mahmood (1997).

Planned replacement is a type of preventive maintenance. Its purpose is to reduce the likelihood of a highly disruptive and expensive failure during operation by investing in a cheaper replacement during scheduled downtime. It is possible to show that, if the failure rate function is constant or decreasing, it is never worthwhile to perform planned replacement. However, if the failure rate function is increasing, as in most practical systems, then there exists a closed form expression for the expected total cost of planned and unplanned replacements. By using spreadsheet analysis, this expression can be minimized to find the optimal replacement time.

The examples discussed here represent only a brief introduction to reliability and maintenance theory. Generalizations that are relevant to robotic systems include multicomponent systems with various other configurations and models that allow various levels of repair. Details can be found in Trivedi (1982). In many cases the analysis is computationally difficult, which has led to the development of reliability analysis software, of which CARE (Stiffler et al., 1979) is an early example. Smith (1993) and Ushakov (1994) each provide a list of commercially available software packages in an appendix. The periodicals *Data Sources* and *Maintenance Technology* provide current information about reliability and maintenance software packages, respectively.

4 SYSTEM OPERATION

In discussing the system operation phase we assume that a robotic system has been chosen and purchased. We further assume that it has been through testing and final installation

and has been declared operational. The next priority is to ensure that the system operates as efficiently as possible. Therefore the focus is on day-to-day operating decisions.

4.1 Issues and Requirements

The overall objectives are to meet production targets and to operate the system smoothly and efficiently. In order to achieve these objectives, several issues and requirements demand the attention of management.

1. *Batch sizes.* Appropriate batch or lot sizes may be determined by setup times, system capacity, and material availability. Decisions need to be made about the size, content, and timing of each batch.
2. *Scheduling.* For each workstation the detailed sequence and timing of jobs must be determined. One of the special features of scheduling in robotic systems is the need to coordinate scheduling and automated material-handling decisions.
3. *Disruptions.* Occasional major disruptions in system operation may occur due to equipment failure, nonavailability of material, or sudden changes in production requirements. It is important to have in place operating procedures for dealing with such situations. Operations research techniques for addressing these problems usually resolve a batching and/or scheduling problem.
4. *Maintenance.* Operating procedures for both scheduled and emergency maintenance should be in place. Some issues in connection with maintenance policies are discussed in Section 3.3.5.

4.2 Performance Measures

The measures used to evaluate system performance at the operation phase are similar to those used at the system design phase. Here, however, they are used with greater detail and precision. This is because the removal of the system design issues makes room for additional considerations within a solvable operations research model, and also because an actual robotic system with known configuration and operating characteristics is being used. Another difference is the time horizon over which performance is being assessed. At the system design phase the only reasonable time horizon is long-term because of the capital cost and disruption involved in changing the system configuration. At the operation phase, however, short-term operating decisions can be made and implemented. The planning process can be repeated when disruptions occur or when production requirements or other parameters change.

4.3 Operations Research Techniques for System Operation

Because short-term operating problems tend to be well defined and present a single objective, they are ideally suited to modeling and solution using operations research techniques. The issues of manufacturing efficiency are not unique to robotic systems. However, there are some characteristics of such systems that distinguish the problems that need to be solved. The following sections emphasize the novel features of using operations research techniques in this environment.

4.3.1 Batching

In traditional manufacturing, *batching* of jobs is frequently necessary because if jobs are performed in the order in which they arrive, there may be many lengthy and expensive setups between them. In batch processing, jobs that are similar are processed together, and within a batch the setup time and cost tend to be small or zero. Only a relatively small number of large setups are necessary when changing batches. Forming a given set of jobs into appropriate batches is a problem that has received considerable attention from manufacturing researchers. Standard references include Wagner and Whitin (1958), Crowston and Wagner (1973), Elmaghraby (1978), Peterson and Silver (1979), Carlson, Jucker, and Kropp (1983), and Afentakis, Gavish, and Karmarkar (1984). Recent studies that integrate batching and scheduling issues include those of Monma and Potts (1989), Albers and Brucker (1993), and Webster and Baker (1995).

However, there are some special characteristics of robotic systems that affect the batching issue. First, to the extent that manufacturing cells are formed around part families of very similar jobs, as discussed in Section 3.3.1, both setup time and cost will typically be much less than in a traditional manufacturing environment. This characteristic tends to reduce the importance of effective batching. Second, there may be unusual constraints

that arise from limited magazine capacities for parts or from the need for different jobs to share specialized tools. These constraints tend to reduce the effectiveness of traditional batching rules in the context of a robotic system. Batching problems that are closely related to robotic systems are discussed by Agnetis, Rossi, and Gristina (1996) and Agnetis, Pacciarelli, and Rossi (1997).

4.3.2 Scheduling

Scheduling concerns the allocation of limited resources to jobs over time. A very readable introduction to this topic is provided by Pinedo (1995). Melnyk (1997) provides a comprehensive survey of scheduling software products used in industry and indicates the suitability of each package for a variety of manufacturing environments. We focus on work directly related to robotic and similar systems. Because this work is extensive, we consider it in two parts, those studies related to robotic cells and those related to other robot-served configurations, respectively. For this purpose, a robotic cell is a robot-served cluster of workstations that contains no internal buffers for work-in-process, and in which only a single family of parts is produced. Thus there are no setup times between parts. The importance of robotic cells is discussed by Levner, Kogan, and Levin (1995), who describe them as the simplest automated blocks, or “islands of automation” in modern computer-integrated manufacturing. Comprehensive surveys of recent work on related scheduling problems are provided by Crama (1997) and Lee, Lei, and Pinedo (1997). Hall and Sriskandarajah (1996) discuss a large variety of scheduling problems with close mathematical connections to robotic systems.

Robotic Cells

An example of a robotic cell is illustrated in Figure 10. An extensive review of the early literature on the scheduling of robotic cells appears in Sethi et al. (1992). More recent references can be found in Hall, Kamoun, and Sriskandarajah (1997). Two types of decisions need to be made in the scheduling of a robotic cell. First, the operations of the material-handling robot must be planned. There are many possible robot moves within the cell. A sequence of moves that returns the cell to its original state forms a *robot move cycle*. As a practical example of a robot move cycle, consider the problem of inserting components into printed circuit boards (see Li and Kumar (1997) for a history of this problem and an alternative solution procedure). In order to identify a robot move cycle

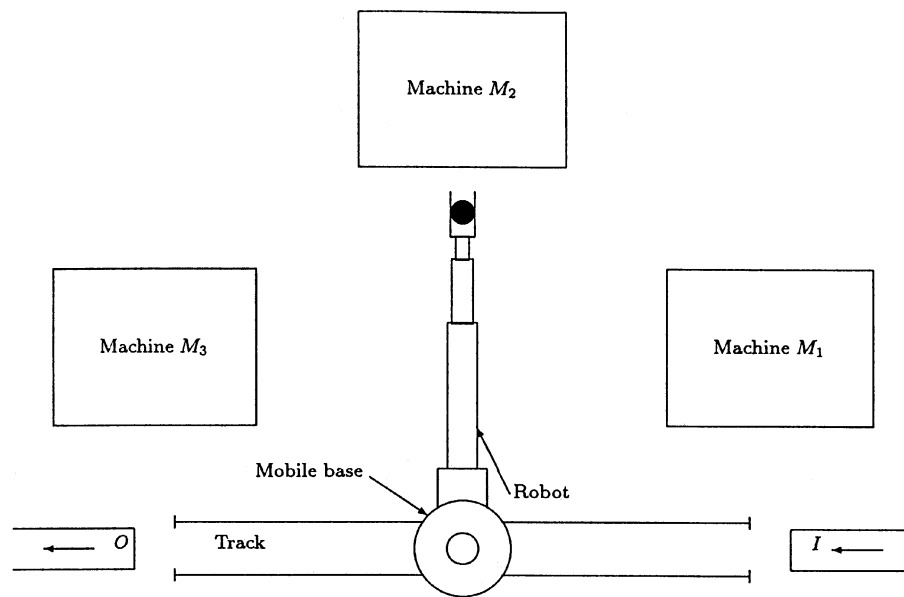


Figure 10 Three-machine robotic cell. (From C. Sriskandarajah, N. G. Hall, and H. Kamoun, “Scheduling Large Robotic Cells Without Buffers,” *Annals of Operations Research* **76**, 1998, 287–321.)

it is first necessary to compute the times between each pair of insertion points. The time used if point j follows point i is computed as the travel time for the robot arm moving from point i to pick up a component, plus the travel time to point j , plus the insertion time at point j . A matrix of these total times can be used to develop an efficient sequence of insertions, which defines a robot move cycle. In robotic cells where several parts may be in a cell simultaneously, it is important to eliminate unnecessary or wasted moves that may make a cycle suboptimal. Sethi et al. (1992) show that in an m machine cell there are $m!$ potentially optimal robot move cycles.

Second, a sequence must be determined for the parts that need to be processed in the cell. Because in general the processing times vary between the parts, the robot move cycle and the part sequence must be determined jointly. Hall et al. (1997) provide an efficient solution procedure that minimizes the time to produce a minimal part set in a two-machine cell. Their procedure has been implemented in a fully equipped two-machine cell at the Manufacturing Systems Laboratory of the University of Toronto. Hall, Kamoun, and Sriskandarajah (1998) and Sriskandarajah, Hall, and Kamoun (1998) identify necessary and sufficient conditions under which an efficient solution procedure exists for larger cells. They show that if these conditions are not satisfied, then the part sequencing problem is intractable (Garey and Johnson, 1979). Variations of this problem are discussed by Kats and Levner (1996).

Robotic cell problems in which all parts are identical have also received the attention of researchers. Sethi et al. (1992) suggest that this problem can be solved by the repetition of a very simple robot move cycle. Hall, Kamoun, and Sriskandarajah (1997) partially prove this result, but do not consider some of the more complex possibilities in their analysis. Crama and van de Klundert (1997) demonstrate that repetition of a simple cycle does provide optimal schedules in three-machine cells. Brauner and Finke (1997) show that a similar result holds in four machine cells with two-part cycles. Furthermore, they show that it does not hold if either the number of machines or the number of parts in a cycle is increased. Levner, Kats, and Levit (1997) and Crama and van de Klundert (1998) provide procedures to minimize cycle time. A variation of this problem is discussed by Levner, Kats, and Sriskandarajah (1996).

Kamoun, Hall, and Sriskandarajah (1998) describe and test heuristic solution procedures for scheduling problems in robotic cells. Levner, Kogan, and Levin (1995) consider a robotic cell consisting of two tandem machines with limited intermediate storage, and provide an efficient procedure for minimizing the schedule length. Levner, Kogan, and Maimon (1995) take into account the effect of transportation time and setups in a similar problem.

Other Configurations

The problems discussed here fall outside our definition of a robotic cell, but nevertheless involve automated material-handling operations that can be performed by robots. As in robotic cells, a major issue is the *coordination* of the material-handling and scheduling functions. However, there may be other relevant issues, such as setup times for jobs, transportation times, time window constraints, and the need to minimize work-in-process. Crane scheduling problems that allow substantial intermediate buffers are considered by Park and Posner (1994, 1996).

Lei and Wang (1991) consider a system involving several workstations in series and two material-handling hoists that move jobs between stations. They develop a procedure for finding the best material-handling cycle that is common to the two hoists. Related work appears in Armstrong, Lei, and Gu (1994). Similar environments in which the objective is to minimize the number of transporters (e.g., robots) are studied by Lei, Armstrong, and Gu (1993) and Armstrong, Gu, and Lei (1996).

Other problems of interest arise in a manufacturing system consisting of several machines, where a job can be processed by any machine. However, job setup on the chosen machine must be performed by a robot. Since there is only one robot, its activity must be coordinated with the schedules of the jobs on the machines. This problem is considered by Glass, Shafransky, and Strusevich (1996) where jobs are dedicated to particular machines and by Hall, Potts, and Sriskandarajah (1998) where jobs can be scheduled on any machine. In both these papers the results include solution procedures, proofs of intractability, and the analysis of heuristics.

Geiger, Kempf, and Uzsoy (1997) describe an interesting application of scheduling methodology to the wet etching process in semiconductor manufacturing. Karuno, Nagamochi, and Ibaraki (1997) consider various scheduling problems that arise in a tree

network served by a single material handler (e.g., a robot). They provide efficient solution procedures for several cases of these problems and distinguish other cases by proving that they are intractable.

4.3.3 Hierarchical Production Planning

The idea of *hierarchical production planning* originated with Hax and Meal (1975). Early studies in this area include those of Bitran, Haas, and Hax (1981) and Graves (1982). The logic behind hierarchical production planning is as follows. Instead of one large planning problem, a number of smaller subproblems are solved, which are typically much easier. Then a (usually approximate) solution to the original problem is constructed from the subproblem solutions. A comprehensive review of the hierarchical production planning literature is provided by Libosvar (1988). Gershwin (1994) gives a very readable introduction to this topic.

Much of the methodology that underlies hierarchical production planning originates with *optimal control theory* (Sethi and Thompson, 1981). For example, Caramanis and Liberopoulos (1992) and van Ryzin, Lou, and Gershwin (1993) use control theory to provide approximate feedback controls for probabilistic manufacturing systems.

However, more recent methodology known as the *hierarchical controls approach* (Lehoczky et al., 1991; Sethi and Zhang, 1994b) may provide an improvement. It has been demonstrated in several probabilistic planning systems that, by using this approach, it is possible to identify solutions that become closer to optimal as the rates of fluctuation of the individual processes within the system increase. The definitive text in this rapidly developing area is by Sethi and Zhang (1994a). Samarutunga, Sethi, and Zhou (1997) provide a computational comparison between a hierarchical controls approach and other approaches by earlier authors to planning a probabilistic manufacturing system. Although the hierarchical controls approach is not designed specifically for automated manufacturing systems, it appears to have the potential to improve their operations.

5 ORGANIZATIONAL DECISION-MAKING AND DECISION SUPPORT SYSTEMS

The preceding sections of this chapter contain a discussion of many operations research techniques that can enhance the design and operation of robotic systems significantly. To use these techniques effectively it is important to understand them in the context of overall *organizational decision-making*. A comprehensive study of organization design theory and practice is provided by Robey (1991). Some interesting perspectives about the subjective implications of organizational structures are offered by Banner and Gagné (1995). Operations research techniques often demonstrate remarkable success at solving well-defined local problems. However, the overall performance of a robotic facility depends upon successful decision-making over different time horizons and at different organizational levels. The means by which the improvements offered by operations research techniques can be translated into overall system performance is a *decision support system*. This system facilitates communication and coordination within the organization. A valuable reference source for background on decision support systems for manufacturing is Parsaei, Kolli, and Hanley (1997). Suri and Whitney (1984) and Lee and Stecke (1996) discuss decision support for automated manufacturing systems. In this section we first discuss the integration of operations research techniques into the decision-making structure of a robotic facility, then provide an overview of a decision support system for such a structure.

An overview of a typical decision-making structure for a robotic facility appears in Table 3. The structure divides into three time horizons, for long-term, medium-term, and short-term decisions, respectively. It should be noted that Sections 2 through 4 above are organized by phase during the *adoption* of a robotic system, whereas all the three time horizons (and corresponding managerial levels) discussed below relate to the *continuing* management and control of a robotic facility. Therefore, there is no one-to-one correspondence between those three sections and the three time horizons of decision-making discussed below.

The first time horizon of decision-making is long-term decisions. These are typically performed by upper-level management over a time horizon of several months or years. Typical long-term decisions include system modification or expansion, major changes in product mix, and the setting of general economic targets. Several relevant and valuable operations research techniques are summarized in the first row, last column of Table 3. Software to support these techniques can be implemented on a workstation computer, for example. Further guidance can be offered by extended part-programming and program-

Table 3 Decision-Making Structure in a Robotic Facility

Time Horizon	Management Level	Typical Tasks	Typical Operations Research Techniques
long-term (months/years)	upper management	system modification or expansion	decision analysis, analytic hierarchy process, queueing models
		product mix changes	linear programming, simulation
		setting of economic targets	forecasting
medium-term (days/weeks)	robotic facility manager	batching tasks	batching
		capacity planning	mixed integer and integer programming, linear programming
		responding to higher- level changes	simulation
short-term (minutes/hours)	robotic facility line supervisor or computer	scheduling operations	scheduling and sequencing
		material-handling and tooling assignments	scheduling and sequencing, heuristics
		problem resolution	simulation

verification tools. Also, the performance of the facility will need to be evaluated, and plans for implementing improvements will need to be developed. Coordination between functional areas of business, e.g., between marketing and product mix decisions, is critical.

The second time horizon is medium-term decisions. These are typically made by a robotic system line manager over a time horizon of several days or weeks. Medium-term decisions include initial batching of production, and planning the availability and effective utilization of resources such as production and material-handling capacity. It may also be necessary to respond to changes in long-term production plans or in material availability. Appropriate operations research techniques to assist in these decisions are shown in the second row, last column of Table 3. Software to support them can be implemented, for example, on a personal computer that is networked to other control functions within the plant.

The third time horizon is short-term decisions. These are typically made by a production system line controller over a time horizon of several minutes or hours. They can also be made automatically as a result of system monitoring by a computer. Short-term decisions include scheduling operations, determining material-handling assignments, resolving temporary bottlenecks and breakdowns, and managing tools and other peripheral equipment. Useful operations research techniques to assist in these decisions are shown in the third row, last column of Table 3. Supporting software can be implemented on a networked personal computer.

As important as decision-making is *within* each level of the organization, so also are the issues of *communication* and *coordination* between levels. Figure 11 illustrates the flow of information between management levels within the organization. Information has to be transferred efficiently from each level to the others, and detailed procedures for such transfers should be established and reviewed regularly. An effective decision support system, as discussed below, is needed to achieve this.

The term decision support system refers to the interaction between human and machine in reaching decisions (Hussain and Hussain, 1981). As discussed by Davis and Hamilton (1993), a decision support system is not intended to replace human decision-makers, but rather to enhance human judgment. Nof (1983) and Lockamy and Cox (1997) provide concise introductions to decision support systems for manufacturing. Within the context of a robotic facility, a decision support system provides a link between the strategic and tactical planning processes illustrated in Figure 11, and supports decision-making for managerial and operational control. The types of information handled by a decision support system in a robotic facility include product and process specifications, order requirements, availability of material, capacity requirements and capabilities, and production

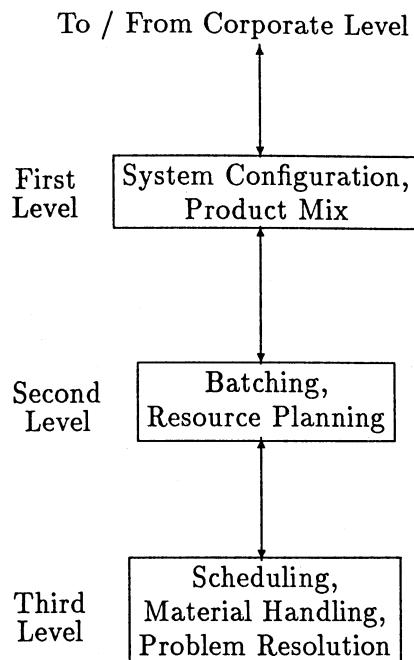


Figure 11 Integration of organizational decision-making in a robotic facility.

schedule details. This information supports strategic decisions about the conversion of resources into outputs and tactical decisions to control the flow of materials. Another important use of this information, as discussed by Hayes, Wheelwright, and Clark (1988), is to provide performance feedback to help with ongoing operational improvements.

An overview of a decision support system for a robotic facility appears in Figure 12. Levary (1997) provides a detailed discussion of a similar system for computer-integrated manufacturing. Some comments about the components of that system now follow. Kaplan (1990) suggests that traditional accounting systems are often not modified sufficiently to allocate costs in a meaningful way in automated manufacturing systems and discusses several alternative approaches. Deterministic and probabilistic operations research techniques are represented as two separate inputs because their role in decision support is conceptually different. Deterministic techniques typically offer a direct (optimal or heuristic) prescription regarding decisions. Probabilistic techniques, by contrast, typically perform an evaluation of a given decision or set of decisions, which can then be compared with similar evaluations for alternative decisions. The former approach offers a much more direct way of identifying good decisions. However, it is not always possible to model a problem deterministically. Because both types of operations research techniques are essentially quantitative and implemented through software, which makes the inclusion of subjective information difficult, that information is represented separately.

There are two distinct databases, an engineering database and a business database. The engineering database includes the characteristics of the product, the robotic facility, and the manufacturing process. Information about the current technology and opportunities for system enhancement is also valuable. The business database includes information on inventory levels, resource availability and production capacity, cash flow, and sales and distribution characteristics. In addition, the accounting system needs to be maintained in the business database. Information from external sources includes, for example, changes in the regulatory environment.

To conclude, appropriate hardware and software components need to be integrated into the organizational decision-making structure in order to maximize the return of a robotic facility. The resulting decision support system should be considered an integral part of the facility, and operation of the facility should explicitly provide for all the components of such a decision support system. It is an effective decision support system that permits

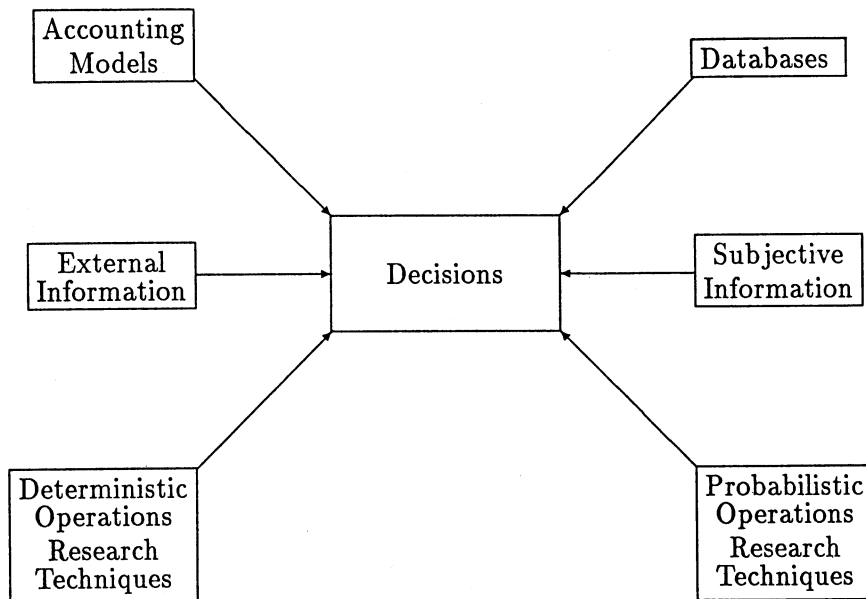


Figure 12 Decision support system in a robotic facility.

the sophisticated techniques of operations research to be fully realized in the performance of a world-class robotic facility.

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